



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

|               |
|---------------|
| JEE(Advanced) |
| PRE-SCORE     |
| 29-10-2023    |

**JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)**

Time : 3 Hours

**PAPER-1 (OPTIONAL)**

Maximum Marks : 198

**IMPORTANT NOTE :** Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

## READ THE INSTRUCTIONS CAREFULLY

### GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

### OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

### DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take  **$g = 10 \text{ m/s}^2$**  unless otherwise stated.

### QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

**SOME USEFUL CONSTANTS**

**Atomic No.** : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

**Atomic masses** : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- |                                    |                                                              |
|------------------------------------|--------------------------------------------------------------|
| • Boltzmann constant               | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$                   |
| • Coulomb's law constant           | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$                   |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$  |
| • Speed of light in vacuum         | $c = 3 \times 10^8 \text{ ms}^{-1}$                          |
| • Stefan-Boltzmann constant        | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$                        |
| • Permeability of vacuum           | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$                |
| • Permittivity of vacuum           | $\epsilon_0 = \frac{1}{\mu_0 c^2}$                           |
| • Planck constant                  | $h = 6.63 \times 10^{-34} \text{ J-s}$                       |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 18)

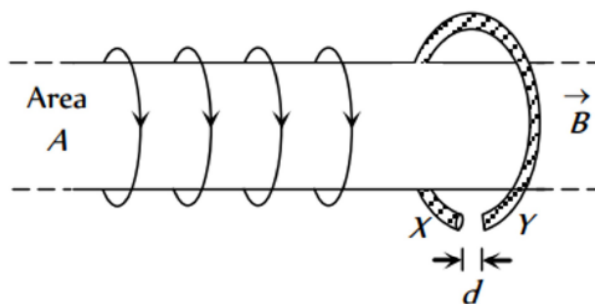
- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A highly conducting ring of radius  $R$  is perpendicular to and concentric with the axis of a long solenoid as shown in fig. The ring has a narrow gap of width  $d$  in its circumference. The solenoid has cross sectional area  $A$  and a uniform internal field of magnitude  $B$ . Now beginning at  $t = 0$ , the solenoid current is steadily increased to so that the field magnitude at any time  $t$  is given by  $B(t) = B + \alpha t$  where  $\alpha > 0$ . Assuming that no charge can flow across the gap, the end of ring which has excess of positive charge and the magnitude of induced e.m.f. in the ring are respectively



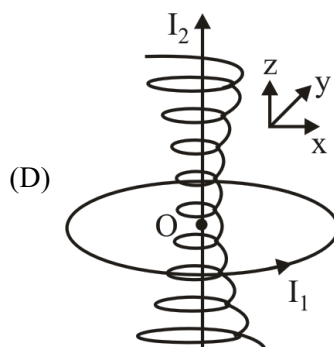
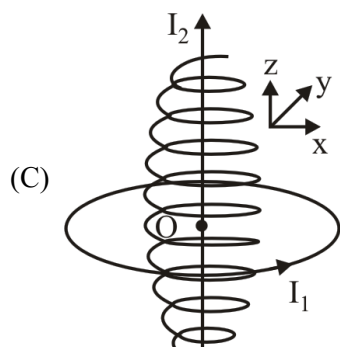
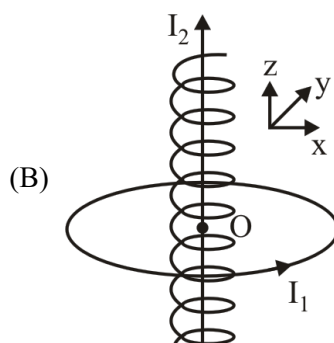
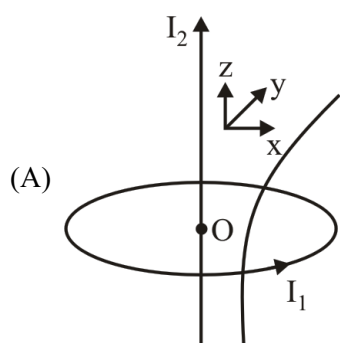
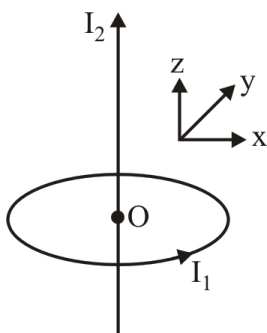
(A) X,  $A\alpha$

(B) X,  $\pi R^2\alpha$

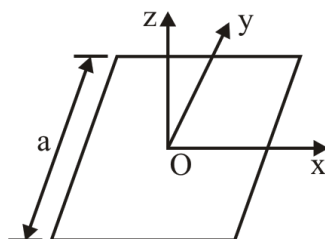
(C) Y,  $A\alpha$

(D) Y,  $\pi R^2\alpha$

2. Which of the following is the correct sketch for a magnetic field line for the current configuration having an infinite wire carrying current  $I_2$  placed along the axis of a coil carrying current  $I_1$ .



3. Square non-deformable superconducting frame having a mass  $m$  and self inductance  $L$  is placed in a non-uniform magnetic field given by  $B_x = -\alpha x$ ,  $B_y = 0$ ,  $B_z = B_0 + \alpha z$ . At  $t = 0$ , the center of the frame is at the origin and sides are parallel to  $x$  axis and  $y$  axis. Acceleration due to gravity is in negative  $z$ -direction. If the frame is released from rest, the variation of  $z$  coordinate as a function of time is :-



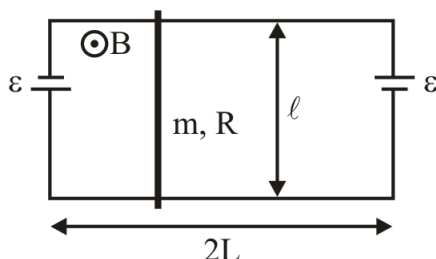
(A)  $z = \frac{mgL}{a^4\alpha^2} \sin\left(\frac{a^2\alpha}{\sqrt{mL}}t\right)$

(B)  $z = \frac{mgL}{a^4\alpha^2} \sin\left(\frac{a^2\alpha}{2\sqrt{mL}}t\right)$

(C)  $z = \frac{mgL}{a^4\alpha^2} \left(1 - \cos\left(\frac{a^2\alpha}{2\sqrt{mL}}t\right)\right)$

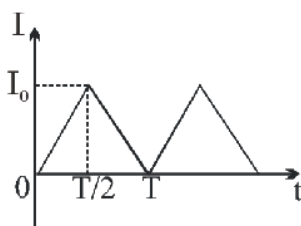
(D)  $z = \frac{mgL}{a^4\alpha^2} \left(\cos\left(\frac{a^2\alpha}{\sqrt{mL}}t\right) - 1\right)$

4. Parallel rails of length  $2L$  are fixed on a horizontal plane at a distance  $\ell$  from each other. Two identical ideal batteries with EMF  $\varepsilon$  are connected to their ends (see figure). On the rails there is a jumper with mass  $m$ , which can progressively slide along them. The whole system is placed in a uniform vertical magnetic field with induction  $B$ . Assuming that the jumper resistance is  $R$ , and the resistance per unit length of each of the rails is equal to  $\rho$ , the period of small oscillations that occur when the jumper is displaced from the equilibrium position.

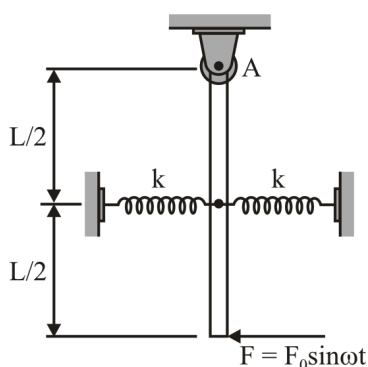


- (A)  $2\pi\sqrt{\frac{mL}{\varepsilon\ell B}}$  (B)  $2\pi\sqrt{\frac{mL(\rho L + R)}{\varepsilon\ell B}}$   
 (C)  $2\pi\sqrt{\frac{2mL(\rho L + R)}{\varepsilon\ell B}}$  (D) None of the above

5. The current- time ( $I-t$ ) graph for triangular wave having peak value  $I_0$  is as shown in figure. The rms value of current is :



- (A)  $I_0$  (B)  $\sqrt{\frac{2}{3}} I_0$  (C)  $\frac{I_0}{\sqrt{2}}$  (D)  $\frac{I_0}{\sqrt{3}}$
6. The uniform rod has a mass of  $m$ . If it is acted upon by a periodic force of  $F = F_0 \sin \omega t$ , determine the angular amplitude of the steady-state vibration.



- (A)  $\frac{F_0}{mg + kL}$  (B)  $\frac{F_0}{2(mg + kL) - m\omega^2 L}$   
 (C)  $\frac{3F_0}{\frac{3}{2}(mg + kL) - m\omega^2 L}$  (D)  $\frac{F_0}{3(mg + kL) - 2m\omega^2 L}$

## SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

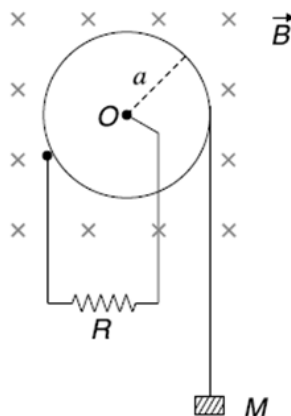
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

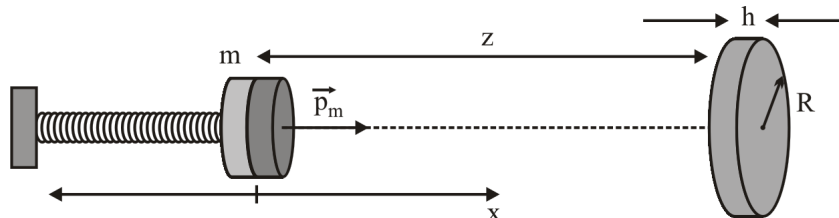
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. A metallic pulley is in the shape of a disc of radius  $a$ . It can rotate freely about a horizontal axis passing through its centre. The moment of inertia of the pulley about this axis is  $I$ . A light string is tightly wrapped around the pulley with its one end connected to a block of mass  $M$ . The centre of the pulley and its circumference are connected to a resistance  $R$  as shown. The contact of resistance at the circumference does not cause any friction. A uniform magnetic field  $B$  is switched on which is parallel to the axis of rotation of the pulley (see Figure). The mass  $M$  is allowed to fall. Assume that resistivity of the material of the pulley is negligible.



- (A) The acceleration of the block of mass  $M$  at the instant its velocity becomes  $v_0$  is  $\frac{Mga^2 - \frac{B^2a^4v_0}{4R}}{I + Ma^2}$ .
- (B) Assuming that the block can fall through a large distance, the terminal speed ( $v_T$ ) that it will acquire is  $\frac{4MgR}{B^2a^2}$ .
- (C) The rate of change of kinetic energy of the pulley at the instant when the speed of the falling block is  $\frac{v_T}{2}$  is equal to  $\frac{IM^2g^2R}{B^2a^2(I + Ma^2)}$ .
- (D) The rate of change of kinetic energy of the pulley at the instant when the speed of the falling block is  $\frac{v_T}{2}$  is equal to  $\frac{IM^2g^2R}{2B^2a^2(I + Ma^2)}$ .

8. Cylindrical magnet of mass  $m$  and magnetic moment  $p_m$ , is attached to the spring of stiffness  $k$  such that it can oscillate along the horizontal axis which is directed along the magnetic moment. At some distance  $z$  from the equilibrium position of the magnet a small metal disc is placed such that its axis coincides with the axis of the magnet. The disk has radius  $R$  and thickness  $h$  ( $h \ll R \ll z$ ), the electrical resistivity of the disk material is  $\rho$ , and the magnetic permeability is equal to  $\mu = 1$ . The magnet is moved from the equilibrium position and starts performing small oscillations. When the speed of magnet is  $v$ ,



- (A) The force exerted by disc on the magnet is equal to  $\frac{9}{32} \frac{\mu_0^2 p_m^2}{\pi \rho z^8} h R^4 v$ .
- (B) The angular frequency of oscillation is  $\sqrt{\frac{k}{m}}$ .
- (C) The rate of power loss in the disk is  $\frac{9 h \mu_0^2 p_m^2 R^4}{32 \pi z^8 \rho} v^2$ .
- (D) The rate of power loss in the disk is  $\frac{9 h \mu_0^2 p_m^2 R^4}{16 \pi z^8 \rho} v^2$ .
9. A wire mesh having identical resistance  $r$  as shown in the figure-1 is placed in a uniform magnetic field in the direction perpendicular to the plane of the mesh. The magnetic field is changing with time such that the rate of change of magnetic flux through one square is  $\left| \frac{d\phi}{dt} \right| = 4\epsilon_0$ . The resistance in wire mesh as shown in figure-1 are replaced by identical capacitance  $C$  as shown in figure-2.

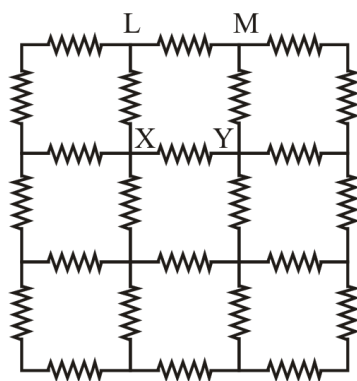


Figure-1

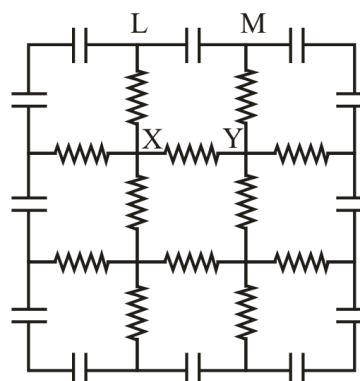
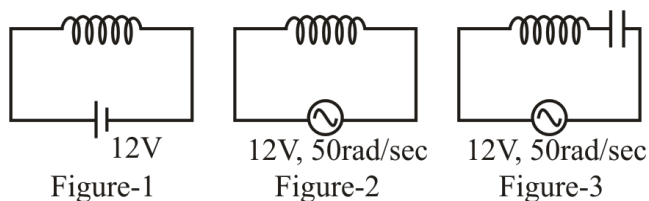


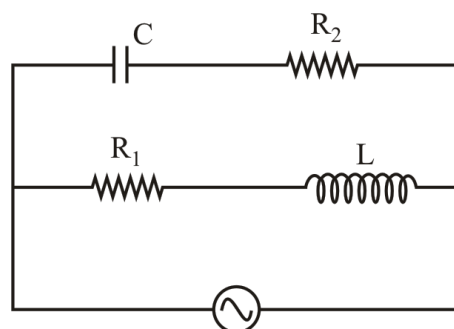
Figure-2

- (A) The current in the resistance  $XY$  in circuit-1 is  $\frac{\epsilon_0}{r}$ .
- (B) The current in the resistance  $LM$  in circuit-1 is  $\frac{7\epsilon_0}{2r}$ .
- (C) The current in the resistance  $XY$  in circuit-2 is  $\frac{\epsilon_0}{4r}$ .
- (D) The charge on the capacitance  $LM$  in circuit-2 is  $5C\epsilon_0$ .

10. A steady current 4 A flows in an inductor coil when connected to a 12V dc source as shown in figure-1. If the same coil is connected to an ac source of 12V, 50 rad/s, a current of 2.4 A flows in the circuit as shown in figure-2. Now after these observations, a capacitor of capacitance  $\frac{1}{50}$  F is connected in series with the coil and with the same AC source as shown in figure-3.

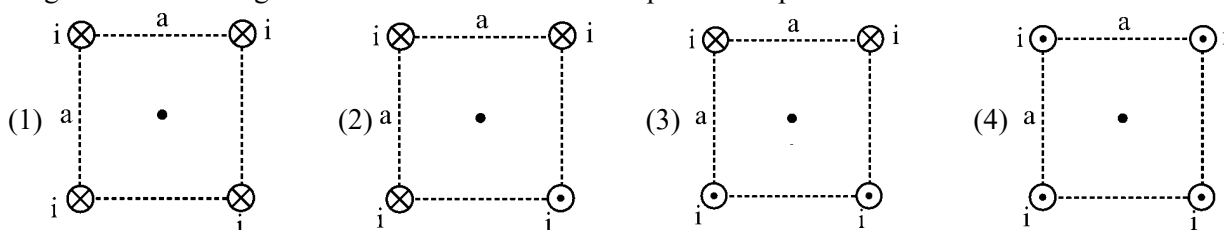


- (A) The inductance of the coil is nearly 0.02 H  
 (B) The inductance of the coil is nearly 0.04 H  
 (C) The average power supplied to the circuit after connecting capacitor in series is 30 W.  
 (D) The average power supplied to the circuit after connecting capacitor in series is 24 W.
11. In the shown circuit  $R_1 = 10\Omega$ ,  $L = \frac{\sqrt{3}}{10}$  H,  $R_2 = 20\Omega$ ,  $C = \frac{\sqrt{3}}{2}$  mF. At the instant when current through  $R_1$  is  $10\sqrt{2}$  A,



$$V = 200\sqrt{2} \sin(100t) \text{ volt}$$

- (A) The current through  $R_2$  is  $5\sqrt{6}$  A.  
 (B) The current through  $R_2$  is zero.  
 (C) The voltage drop across the capacitance is  $200\sqrt{2}$  V  
 (D) The voltage drop across the inductance is  $100\sqrt{6}$  V.
12. Four long wires, with identical currents, either directly into or directly out of the page, are placed at four corners of a square (of side  $a$ ) in four different arrangement as shown.  $B_1$ ,  $B_2$ ,  $B_3$  and  $B_4$  are the magnitude of net magnetic field at the center of the square in respective cases.



- (A)  $B_1 = B_4 = 0$                       (B)  $B_2 = \frac{\sqrt{2}\mu_0 i}{\pi a}$                       (C)  $B_3 = \frac{2\mu_0 i}{\pi a}$                       (D)  $B_2 = B_3$

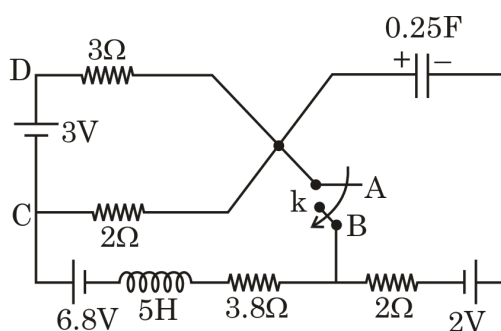
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

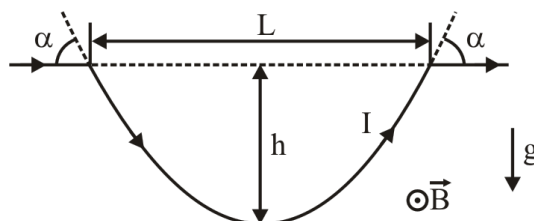
*Full Marks* : +4 If ONLY the correct numerical value is entered.

*Zero Marks* : 0 In all other cases.

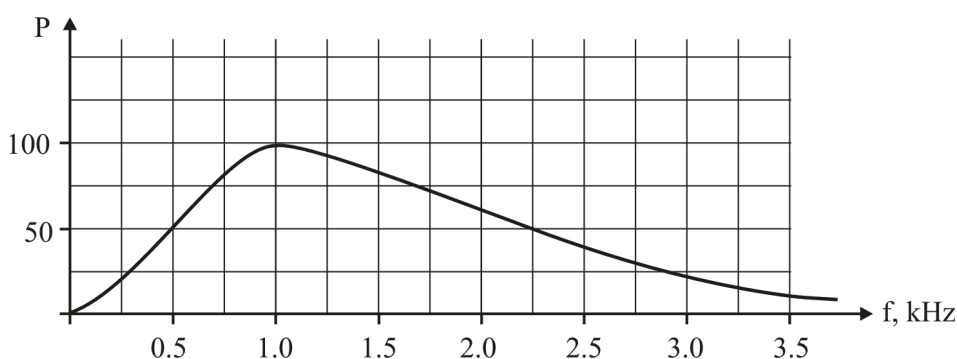
1. A deflection of 24 divisions on a ballistic galvanometer (for ballistic galvanometer charge  $\propto$  deflection) is obtained by discharging through it a capacitor of  $3 \mu\text{F}$  charged to a potential difference of 2 V. When the same galvanometer is connected to a coil of 80 turns, each of diameter 1 cm, and the coil is quickly introduced into a strong magnetic field, a deflection of 16 divisions is observed. Calculate the strength of the magnetic field. The total resistance of the galvanometer circuit is  $2000 \Omega$ .
2. In order to impart an angular velocity to an earth satellite, the geomagnetic field can be used. Find the maximum possible angular velocity (in  $10^{-2} \text{ rad/s}$ ) about its own axis gained by the satellite if a storage battery with a capacity of  $Q = 5 \text{ Amp-hours}$  is discharged suddenly through a coil of  $N = 20$  turns wound around the satellite's surface along the circumference of the largest circle. The satellite has a mass of  $m = 10^3 \text{ kg}$  and is a thin walled uniform sphere. The geomagnetic field is parallel to the winding plane and its flux density is  $B = 0.5 \text{ Gauss}$ .
3. In the given circuit, initial charge on the capacitor is 1.5 C with polarity as shown in figure and current in the inductor is zero. Now at  $t = 0$  if the key  $k$  is closed, then the minimum current through the key is :-



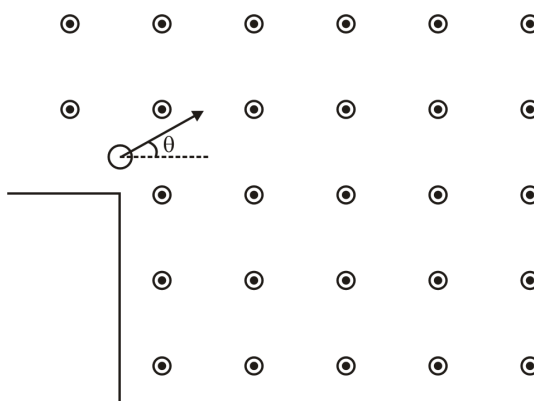
4. A wire of mass  $m$  is suspended from two points on the same horizontal level as shown in the diagram. The wire is placed in a uniform horizontal magnetic field and a current  $I$  flows through it. The tension (in N) at the lowest point of the wire is ( $m = 1 \text{ kg}$ ,  $I = 2 \text{ A}$ ,  $L = 2 \text{ m}$ ,  $B = 3 \text{ T}$ ,  $\alpha = 37^\circ$ ,  $H = 3 \text{ m}$ )



5. The graph below shows the variation of power as a function of frequency of the AC source in a series RLC circuit. The quality factor for the circuit is.



6. Professor Proton has discovered a new planet on one of his planetary expeditions. He wants to measure the magnetic field of the planet he has found. Professor Proton has brought all the necessary equipment required to carry out the following experiment. A proton is launched off a large cliff at a speed  $v$  and an angle  $\theta = 30^\circ$  with respect to the horizontal plane at the magnetic equator of the planet. The magnetic field acting perpendicularly on the particle can be assumed to be perfectly horizontal and coming out of the page, as shown in the diagram. How strong is the magnetic field at the magnetic equator of this planet if the time period of oscillation of horizontal component of velocity of the particle is  $5 \times 10^{-5} \text{ s}$ ? Write your answer in terms of  $\mu\text{T}$  (micro-Teslas). Take mass of proton  $= 1.6 \times 10^{-27} \text{ kg}$ , charge on proton  $= 1.6 \times 10^{-19} \text{ C}$  and  $\pi = 3.14$ .



**PART-2 : CHEMISTRY**  
**SECTION-I (i) : (Maximum Marks: 18)**

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

*Full Marks* : +3 If **ONLY** the correct option is chosen.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered)

*Negative Marks* : -1 In all other cases

1. At what minimum concentration of  $\text{OH}^-$  ions will  $10^{-3}$  mole of  $\text{Zn(OH)}_2$  go into solution as  $\text{Zn(OH)}_4^{2-}$  in 1 L aqueous solution.  
 $\text{Zn(OH)}_2(\text{s}) + 2\text{OH}^-(\text{aq.}) \rightleftharpoons \text{Zn(OH)}_4^{2-}(\text{aq.})$  ( $K_C = 10^{-2}$ )  
 (A) 0.1 M (B) 0.632 M (C) 0.0316 M (D) 0.316 M
2. A aqueous solution is a mixture of 0.06 M KCl and 0.06 M KI. Solid  $\text{AgNO}_3$  is being added gradually till AgCl starts precipitating ( $K_{\text{sp}} [\text{AgCl}] = 1 \times 10^{-10}$  and  $K_{\text{sp}} [\text{AgI}] = 4 \times 10^{-16}$ ). The concentration of iodide ion at this instant will be nearly equal to :  
 (A)  $4.0 \times 10^{-5}$  M (B)  $4 \times 10^{-8}$  M (C)  $2.4 \times 10^{-8}$  M (D)  $2.4 \times 10^{-7}$  M
3. The degree of dissociation of water in a 0.1 M aqueous solution of HCl at a certain temperature  $t^\circ\text{C}$  is  $3.6 \times 10^{-15}$ . The temperature 't' must be : [Take density of water at  $t^\circ\text{C} = 1 \text{ g/mL}$ ]  
 (A)  $< 25^\circ\text{C}$  (B)  $= 25^\circ\text{C}$   
 (C)  $> 25^\circ\text{C}$  (D) Insufficient data to predict
4. Two moles of an ideal gas ( $\gamma = 3/2$ ) are made to expand reversibly and adiabatically to 4 times its initial volume. The change in entropy of the system during expansion is : (Given :  $R = 2 \text{ cal/K/mole}$ ,  $\log_{10} 2 = 0.3$ ,  $\log_{10} 3 = 0.48$ )  
 (A) 5.6 cal/k (B) 11.2 cal/K (C) 2.8 cal/K (D) None of these
5. 0.5 mole each of two ideal gas A ( $C_V = R$ ) and B ( $C_V = R$ ) are taken in a container and expanded reversibly and adiabatically from  $V = 1$  litre to  $V = 2$  litre starting from initial temperature  $T = 300 \text{ K}$ .  $\Delta H$  for the process (in cal/mol) is :  
 (A) -500 cal (B) -900 cal (C) -450 cal (D) -600 cal
6. Calculate  $\Delta G$  for the given phase transition :  
 $\text{H}_2\text{O} (\ell, 2 \text{ mole}, 1 \text{ atm}, 100^\circ\text{C}) \rightarrow \text{H}_2\text{O} (\text{g}, 2 \text{ mole}, 2 \text{ atm}, 100^\circ\text{C})$   
 Assume  $\text{H}_2\text{O}(\text{g})$  to behave as an ideal gas. Give your answer in Kcal rounded off to nearest integer.  
 (A) 3 (B) 2 (C) 1 (D) 4

## SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

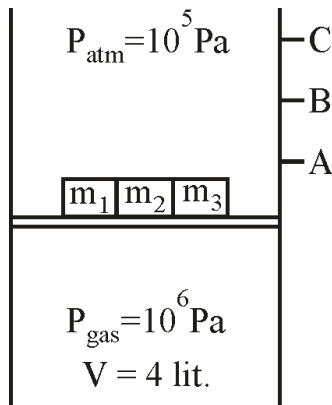
7. Select correct statements :

- (A) For an ideal gas (undergoing no association or dissociation) adiabatic free expansion & isothermal free expansion have equal change in enthalpy for same volume change  $V_1$  to  $V_2$  starting from same initial state.
- (B) An ideal mono atomic gas follows the process  $P^5V^7 = \text{constant}$ . If heat is withdrawn from the gas during this process, then temperature of gas must increase.
- (C) For any cyclic process :  $\int \frac{dq}{T} \leq 0$
- (D) Adiabatic process is always an isentropic process.

8. Which of the following is/are incorrect statement(s) about  $10^{-3}$  M aqueous solution of HCl at 298 K ?
- (A) On dilution pH will decrease
- (B)  $[\text{H}_3\text{O}^+]$  from  $\text{H}_2\text{O}$  is nearly  $10^{-3}$  M
- (C)  $\alpha$  for  $\text{H}_2\text{O}$  is nearly  $1.8 \times 10^{-13}$
- (D)  $[\text{OH}^-]$  in solution is nearly  $10^{-12}$
9. A 40.0 ml solution of weak base, BOH is titrated with 0.1 N HCl solution. The pH of the solution is found to be 10.0 and 9.0 after adding 5.0 ml and 20.0 ml of the acid, respectively. The correct informations is/are : ( $\log 2 = 0.3$ )
- (A)  $K_b = 2 \times 10^{-5}$
- (B)  $\text{p}K_b = 4.5$
- (C) Initially :  $[\text{BOH}] = \frac{3}{40} \text{ M}$
- (D) pH at equivalence point  $< 7$ .
10. Choose the incorrect statement(s) among the following options :
- (A) Change in Gibb's function at constant temperature and Pressure in any process involving no non-PV work is negative.
- (B) For a reaction whose standard entropy change is positive, as temperature is increased, equilibrium constant for the reaction increases necessarily.
- (C) Exothermic reaction can not be spontaneous at higher temperature.
- (D) Magnitude of change in Gibb's function at constant temperature and pressure is equal to magnitude of maximum non-PV work that can be done by the system.

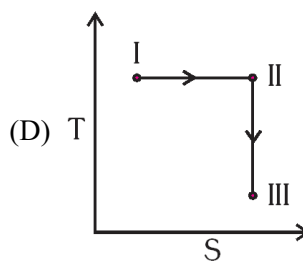
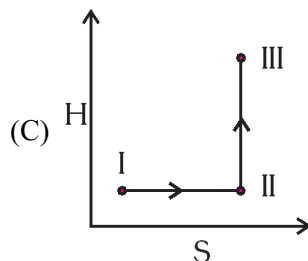
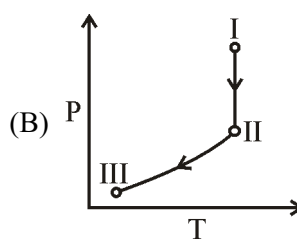
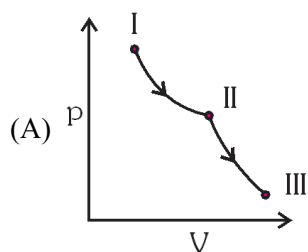
11. Consider an isothermal cylinder massless piston assembly in which ideal gas is filled. Cross sectional area of the cylinder =  $1\text{ m}^2$ . The masses  $m_1$ ,  $m_2$  and  $m_3$  are kept on the piston. When  $m_1$  is removed, piston moves upto point A. When  $m_1$  &  $m_2$  both are removed piston moves upto point B and when  $m_1$ ,  $m_2$  and  $m_3$  all three are removed, piston moves upto point C.

[Given :  $m_1 = 2 \times 10^4 \text{ kg}$ ,  $m_2 = 3 \times 10^4 \text{ kg}$ ,  $g = 10 \text{ m/s}^2$ ]



- (A) ' $m_3$ ' =  $5 \times 10^4 \text{ kg}$   
 (B) Volume of gas at point B is 10 litre  
 (C) Pressure of gas at point B is  $5 \times 10^5 \text{ Pa}$   
 (D) Work done when piston moves from B to C is  $-3200 \text{ J}$
12. An ideal gas undergoes a reversible isothermal expansion from state I to state II followed by a reversible adiabatic expansion from state II to state III. The correct plot(s) representing the changes from state I to state III is(are)

(p : pressure, V : volume, T : temperature, H: enthalpy, S : entropy)



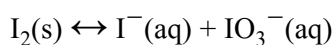
## SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If ONLY the correct numerical value is entered.

*Zero Marks* : 0 In all other cases.

1. Calculate pH at equilibrium of the solution in which following reaction taking place under basic medium at 300 K.



Given

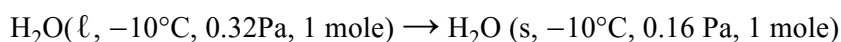
| Substance                                     | $\text{I}^-(\text{aq})$ | $\text{IO}_3^-(\text{aq})$ | $\text{H}_2\text{O}(\ell)$ | $\text{OH}^-(\text{aq})$ |
|-----------------------------------------------|-------------------------|----------------------------|----------------------------|--------------------------|
| $\Delta G^\circ_f$ (in $\text{KJ mol}^{-1}$ ) | -50                     | -123.5                     | -233                       | -150                     |

Equilibrium concentration of  $\text{I}^- = 0.1 \text{ M}$

Equilibrium concentration of  $\text{IO}_3^- = 0.1 \text{ M}$

$$R = \frac{25}{3} \text{ J mol}^{-1} \text{ K}^{-1}; \ln 10 = 2.3$$

2. The vapour pressures of liquid water and ice at  $-10^\circ\text{C}$  are 0.32 and 0.16 Pa, respectively. What is the magnitude of Gibbs free energy change for the process (in Joule).



[Given :  $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$ ,  $\ln 2 = 0.7$ ]

(Answer to the nearest integer)

3.  $2\text{A}(\text{s}) \rightleftharpoons \text{B}(\text{g}) + 2\text{C}(\text{g}) + 3\text{D}(\text{g})$

Total pressure developed in a closed container by decomposition of A at equilibrium is 12 bar at  $727^\circ\text{C}$ .

Calculate  $|\Delta G^\circ|$  (in kcal) of the reaction at  $727^\circ\text{C}$  ( $R = 2 \text{ cal/mole-K}$ ,  $\ln 2 = 0.7$ ,  $\ln 3 = 1.1$ )

4. When 1000 bar constant external pressure is applied on FCC crystal of metal 'M' then its crystalline structure is changed from FCC to BCC. If radius of metal is  $100\sqrt{3}\text{\AA}$  and initially in the FCC lattice there were  $10^{15}$  unit cells, then calculate work done by the metal crystal, in joule. [Take  $\sqrt{6} = 2.5$ ]
5.  $\text{O}_2$  is kept in a vessel at 300 K. Following equilibrium is established  $3\text{O}_2(\text{g}) \rightleftharpoons 2\text{O}_3(\text{g})$ . At equilibrium partial pressure of  $\text{O}_2 = 10^5$  bar and partial pressure of  $\text{O}_3 = 10^{-10}$  bar. Calculate  $\Delta_r G^\circ$  (in Kcal) of the above reaction.  
[Given :  $R = 2 \text{ cal K}^{-1} \text{ mol}^{-1}$ ;  $\log_e 10 = 2.3$ ]
6. 500 ml of an aqueous solution of 0.01 M  $\text{AgNO}_3$  is mixed with 250 ml of aqueous solutions of each of NaBr and NaCl, each having molarity 0.02 M. The equilibrium concentration of  $\text{Br}^-$  when AgCl starts precipitating is  $x \times 10^{-y}$  mole/L. The value of 'x' is :  
[Given  $K_{\text{sp}}(\text{AgBr}) = 5 \times 10^{-13}$ ;  $K_{\text{sp}}(\text{AgCl}) = 10^{-10}$ ].

**PART-3 : MATHEMATICS**  
**SECTION-I (i) : (Maximum Marks: 18)**

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

*Full Marks* : +3 If **ONLY** the correct option is chosen.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered)

*Negative Marks* : -1 In all other cases

- 
1. Two triangles ABC and DEF are formed by three tangents and the corresponding three normals to a parabola  $y^2 = 4ax$  respectively. The orthocenter of  $\triangle ABC$  has coordinates  $(-5, 4)$ . Then, distance of orthocenter of  $\triangle DEF$  from axis of parabola is
- (A) 5 units  
(B) 2 units  
(C) 4 units  
(D) cannot be determined
2. Let P be any point on a directrix of an ellipse of eccentricity  $e$ , S be the corresponding focus and C is the centre of the ellipse. The line PC meets the ellipse at A and the angle between PS and tangent at A is  $\alpha$ , then  $\alpha$  is equal to -
- (A)  $\tan^{-1}e$  (B)  $\frac{\pi}{2}$   
(C)  $\tan(1 - e^2)$  (D) none of these
3. For an ellipse  $\frac{x^2}{9} + \frac{y^2}{4} = 1$ , with vertices A & A', tangent drawn at the point 'P' in the first quadrant meets the y-axis in Q and the chord A'P meets the y-axis in M. If 'O' is the centre of ellipse, then  $OQ^2 - MQ^2$  is equal to
- (A) 9 (B) 13  
(C) 4 (D) 5

4. The normal at P to a hyperbola of eccentricity  $e$ , intersects its transverse and conjugate axes at L and M respectively. If locus of the mid-point of LM is hyperbola, then eccentricity of the hyperbola is :
- (A)  $\left(\frac{e+1}{e-1}\right)$  (B)  $\frac{e}{\sqrt{e^2-1}}$   
 (C)  $e$  (D) None of these
5. Let AB and CD are two chords of hyperbola  $9x^2 - y^2 = 1$ . If AB and CD are perpendicular and S(1,0) is their point of intersection, then the value of  $\frac{1}{(AS)(BS)} + \frac{1}{(CS)(DS)}$  is equal to
- (A) 1 (B) 8  
 (C) 9 (D)  $\frac{1}{9}$
6. The locus of point of intersection of tangents drawn at ends of all normal chords to parabola  $y^2 = 8x - 8$  is -
- (A)  $y^2(x-3) = -32$  (B)  $y^2(x+3) = -32$   
 (C)  $x^2(y+3) = -32$  (D)  $x^2(y-3) = -32$

## SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

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*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

- 
7. A tangent is drawn at any point  $P(t)$  on the parabola  $y^2 = 8x$  and a point  $Q(\alpha, \beta)$  is taken on it from which pair of tangents  $QA$  and  $QB$  are drawn to the circle  $x^2 + y^2 = 4$ , where  $A$  &  $B$  are points of contact. Now if point  $P(t)$  varies on the parabola then the locus of all such point of concurrency of the chord of contact  $AB$  is denoted by  $S = 0$ , then -

(A) Equation of  $S = 0$  is  $2x + y^2 = 0$

(B) Equation of  $S = 0$  is  $x + y^2 = 0$

(C) Area bounded by  $S = 0$  and the line  $2x + 1 = 0$  is  $\frac{2}{3}$  sq. units

(D) Tangents drawn at the extremities of intersection of  $S = 0$  and the line  $2x - y + 1 = 0$  to the curve  $S = 0$  includes an angle  $75^\circ$ .

8. Let  $P(x_0, y_0)$  be any point on the parabola  $y^2 = 4x$ , if normal at P intersect the parabola again at Q and tangents at P and Q intersect at point R -
- (A) acute angle between normal at P and tangent at Q to the parabola is  $\theta$ , then  $\tan \theta = \frac{|y_0|}{4}$
- (B) length of normal chord PQ is  $\frac{4(x_0 + 1)^{3/2}}{x_0}$
- (C) ordinate of point R is  $-\frac{4}{y_0}$
- (D) ordinate of circumcentre of triangle PQR is  $\frac{-2(x_0 + 3)}{y_0}$
9. Let  $S_1$  is a hyperbola and  $S_2$  be an ellipse whose centers are at origin. The transverse axis of  $S_1$  and major axis of  $S_2$  lie along the x-axis and y-axis respectively. Let H be the hyperbola  $x^2 - y^2 = 7$ . The straight line  $4x - 3y = 7$  touches the curves H,  $S_1$  and  $S_2$  at P, Q and R respectively. Suppose  $PQ = \frac{5}{2} = QR$ . If  $e_1$  and  $e_2$  are eccentricities of  $S_1$  and  $S_2$  respectively. Then -
- (A)  $e_1 = \frac{23}{15}$  (B)  $e_2 = \frac{1}{2}$
- (C)  $15e_1^2 + 4e_2^2 = 24$  (D)  $e_1$  is an irrational number
10. Let H :  $y(3y + 4x) = -4$  is a hyperbola and  $y = mx + c$  is its conjugate axis. Length of latus rectum of H is L, eccentricity e and  $(x_1, y_1)$  is one vertex with  $y_1 > 0$ , then -
- (A)  $m = 2$  (B)  $L = 1$  (C)  $4e^2 = 5$  (D)  $5y_1^2 = 4$
11. Let two parabolas have a common axis where focus of each being exterior to the other and let  $\ell_1, \ell_2$  be their latus rectums then the locus of the mid points of the intercepts between the parabolas made on the lines parallel to the common axis is a
- (A) Straight line if  $\ell_1 = \ell_2 \forall \ell_1, \ell_2 \in \mathbb{R}^+$
- (B) Parabola if  $\ell_1 \neq \ell_2 \forall \ell_1, \ell_2 \in \mathbb{R}^+$
- (C) Parabola  $\forall \ell_1, \ell_2 \in \mathbb{R}^+$
- (D) Hyperbola if  $\ell_1 = 2\ell_2 \forall \ell_1, \ell_2 \in \mathbb{R}^+$
12. A parabola whose focus is S(6, 8) is touching the co-ordinate axes, then which of the following statement(s) is/are correct ?
- (A) The equation of circle whose diameter is the portion of the tangent at the vertex of the parabola between the co-ordinate axis is  $x^2 + y^2 - 6x - 8y = 0$ .
- (B) If P and Q are the ends of a focal chord of the parabola, then  $\frac{1}{SP} + \frac{1}{SQ} = \frac{5}{24}$ .
- (C) Equation of the latus rectum is  $4x + 3y = 48$ .
- (D) If P and Q are the ends of a focal chord of the parabola, then  $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{5}$ .

## SECTION-II : (Maximum Marks: 24)

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- Answer to each question will be evaluated according to the following marking scheme:

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1. Chord of the curve  $3x^2 - y^2 - 2x + 4y = 0$  which subtends a right angle at the origin, always passes through vertex of a concave up parabola whose axis is parallel to y-axis and length of latus rectum is 8. If extremity of latus rectum of parabola is  $L_1(\alpha, \beta)$  and  $L_2(\gamma, \delta)$ , then find the value of  $(\alpha + \beta + \gamma + \delta)$  ?
2. One of the focus of ellipse  $\frac{x^2}{8} + \frac{y^2}{4} = 1$  is at S and P is the upper end of its minor axis. A parabola is drawn whose vertex is at origin and P, S are extremities of its latus rectum, then one of its possible equations is of the form  $\lambda(x-1)^2 + \lambda(y-1)^2 = (x+y+\lambda)^2$ , where  $\lambda$  is
3. From a point P on parabola  $y^2 = 4x$  a normal is drawn to intersect it at Q and tangent drawn at P intersect x-axis at R. If the locus of centroid of  $\Delta PQR$  is  $108xy^2 = (a+by^2)^2$ , then the value of  $(b-a)$  is
4. Consider an ellipse E :  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  and a parabola P :  $y^2 = 4c(x+a)$ , ( $a > 0$ ). A straight line intersects 'E' at points A and B and 'P' at points C and D. If the angle subtended by AB at the centre of the ellipse is  $90^\circ$  then the locus of point of intersection of tangents to the parabola 'P' at C and D is  $a^2b^2(y^2 + \alpha c^2) = 4c^2(x + \beta a)^2(a^2 + b^2)$ . The value of  $\alpha$  is
5. Let  $(-1, -1)$  be the focus of a parabola and  $3x - y = 8$  be the tangent drawn to it at point  $(7, 13)$  on the parabola. If  $ax + by + 19 = 0$  is the equation of directrix of the parabola then the value of  $a + b$  is (where  $a, b \in \mathbb{N}$ )
6. Consider an ellipse  $4x^2 = 52 - 13y^2$  and a variable point P on the line  $y + 2x = 12$  such that  $\angle F_1PF_2$  is maximum where  $F_1$  and  $F_2$  are the foci of the given ellipse then  $\left(\frac{F_2P}{F_1P}\right)^2$  is equal to (where  $F_1$  lies on positive x-axis)

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. ....

I have read all the instructions  
and shall abide by them.

\_\_\_\_\_  
Signature of the Candidate

I have verified the identity, name and Form  
number of the candidate, and that question  
paper and ORS codes are the same.

\_\_\_\_\_  
Signature of the Invigilator

**Space for Rough Work**

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E-24/24

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